

Yuer Gao

Curriculum Vitae

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RESEARCH SUMMARY

Research Focus: Marine robotics and autonomy, with emphasis on disturbance-aware motion planning, robust perception, and ROS-based real-robot systems for underwater navigation and inspection.

Selected Contributions:

- **Real-Time Underwater State Estimation** (RA-L 2025): Developed a tightly coupled vision–optical fusion framework with latent-variable motion models, achieving **5.65 mm** position RMSE at **62 FPS** and improving accuracy by **6.4×** over a vision-only baseline.
- **Robust Monocular Localization** (CASE 2025): Proposed **MULoc**, a monocular localization framework fusing RGB and geometric depth cues, achieving **92%** tracking success under turbidity and **< 1%** trajectory-length localization error at **25 FPS**.
- **Disturbance-Aware Marine Motion Planning** (TMECH, under review): Designed a redundancy-resolving planner for **8-thruster** ROVs, reducing target-region particle velocity by **67%** and improving reconstruction RMSE by **55%** (**1.9 mm** vs. **4.3 mm**) relative to disturbance-unaware baselines.

RESEARCH EXPERIENCE

The Hong Kong University of Science and Technology (GZ) 2023 – Present (PhD Candidate)

Underwater Autonomy, Perception, and Disturbance-Aware Reconstruction

- Developed a disturbance-aware shared-autonomy framework for over-actuated ROVs that exploits actuation redundancy to suppress sediment resuspension and thruster-induced turbulence.
- Built and validated ROS-based perception-planning-control pipelines for underwater inspection and reconstruction, integrating state estimation, localization, and motion planning modules on real robotic platforms.
- Conducted **440** real-robot trials, achieving **98.5%** reconstruction success and **1.9 mm** RMSE, demonstrating robust operation under visually degraded and hydrodynamically challenging conditions.
- Studied actuation–perception coupling in underwater robotic systems, to enable reliable autonomy for marine inspection and adaptive sensing.

Xi'an University of Technology 2021 – 2022 (Lead Researcher)

Underwater Vehicle–Manipulator System (UVMS) Research

Worked on modeling, motion planning, and obstacle avoidance for UVMS in marine environments, and developed a ROS-based simulation platform for autonomous navigation and trajectory optimization.

TECHNICAL SKILLS

Robotics: ROS/ROS2, RViz, Gazebo, MoveIt, robot simulation, system integration

Programming: C/C++, Python, MATLAB, Linux

Perception & Vision: Visual localization, multi-modal fusion, underwater perception, YOLO

Planning & Control: Motion planning, trajectory optimization, disturbance-aware autonomy

Hardware / Communication: STM32, DSP, UART, I2C, CAN, RS422/RS232, TCP/IP

EDUCATION

- 2023 – 2026** **PhD in Smart Manufacturing**
The Hong Kong University of Science and Technology (GZ)
Expected: Dec. 2026 • GPA: 3.5/4.0
- 2020 – 2023** **Master of Control Science and Engineering**
Xi'an University of Technology
GPA: 4.4/5.0
- 2017 – 2021** **Bachelor of Engineering in Automation**
Xi'an University of Technology
GPA: 4.2/5.0 • Top 3%

SELECTED AWARDS

- 2024** **Model Mark — Third Prize** HKUST(GZ), Team Lead
- 2022** **National Scholarship (Graduate)** (twice)
- China Robotics & AI Competition** (1st Prize)
- 2021** **Internet+ Innovation** (Bronze/Gold)

ACADEMIC SERVICE

Journal & Conference Reviewer

- Ocean Engineering
- IEEE International Conference on Robotics and Automation (ICRA)
- IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)

Teaching Assistant

- UCMP 6010: Cross-disciplinary Research Methods (HKUST)

Academic Mentoring / Evaluation

- **Evaluator / Mentor**, Red Bird Challenge: Selected and evaluated prospective MPhil students during a two-week intensive selection camp at HKUST(GZ).

LEADERSHIP

- 2024** **Model Mark 2024 Competition**
- Team Leader / Organizer** Represented HKUST(GZ) as a team leader to coordinate and compete in the Model Mark competition. Awarded the **Third Prize** for innovative modeling and technical execution.

SELECTED FIRST-AUTHOR PUBLICATIONS

Y. Gao, T. Xu, Y. Cai. “Real-Time Millimeter-Accurate Underwater Pose Estimation via Tightly-Coupled Fusion of Vision and Optical Tracking.” *IEEE Robotics and Automation Letters (RA-L)*, 2025.

Y. Gao, T. Xu, Q. Liu, Y. Cai. “Disturbance-Aware Motion Planning for Over-Actuated Underwater Vehicles Exploiting Actuation Redundancy for High-Fidelity 3D Reconstruction.” *IEEE/ASME Transactions on Mechatronics (TMECH)*. Under review

Y. Gao, Y. Cai. “Bidirectionally Coupled Global–Local Path Planning for 3D Underwater Navigation in Ocean Currents.” *IEEE Journal of Oceanic Engineering (JOE)*. Under review

Y. Gao, Y. Cai. “MULoc: Robust Monocular Visual Localization for Underwater Robots via Multi-modal Feature Fusion.” *IEEE International Conference on Automation Science and Engineering (CASE 2025)*.

Y. Gao, Y. Zhao, Y. Cai. “CoralVLM: A Task-Oriented Vision-Language Benchmark Toward Underwater Embodied Coral Reef Inspection.” *IEEE Robotics and Automation Letters (RA-L)*. Submitted

Y. Gao, J. Luo, Y. Cai. “Operator-in-the-Loop Underwater Coral Inspection: Instance Grounding, Active 3D Anchoring, and Disturbance-Aware Shared Autonomy.” *IEEE Transactions on Automation Science and Engineering (TASE)*. Under review

Y. Gao, X. Zhang, et al.. “UVMS Task-Priority Planning Framework for Underwater Task Goal Classification Optimization.” *Frontiers in Neurobotics*, 2022.

Y. Gao, T. Xu, and Y. Cai. “Minimum-Snap Trajectory Optimization for Underwater Manipulator.” *Proceedings of the 2024 International Conference on Advanced Robotics and Mechatronics (ICARM)*, 2024.

SELECTED CO-AUTHOR PUBLICATIONS

Z. Liao, W. Wei, L. Zhang, **Y. Gao**, Y. Cai. “An Augmented Reality-Enabled Digital Twin System for Reconfigurable Soft Robots: Visualization, Simulation and Interaction.” *Computers in Industry*, 2025.

Q. Liu, T. Xu, F. Zhan, **Y. Gao**, Z. Liao and Y. Cai. “Autonomous Magnetically Reconfigurable Soft Gripping System for Adaptive Manipulation in Smart Manufacturing.” *Computers & Industrial Engineering*, 2026.

T. Xu, **Y. Gao**, and Y. Cai. “AI-Based Structural Feature Recognition for Robotic Grasping of Additive Manufacturing Parts.” *Proceedings of the International Conference on Flexible Automation and Intelligent Manufacturing (FAIM)*, 2024.

Y. Gao, X. Zhang, **Y. Gao**, T. Shang, and Q. Yang. “Underwater Manipulator Control Based on Neural Network and Fuzzy Compensation.” *Computer Engineering and Applications*, vol. 58, no. 15, pp. 317–323, 2022. (Chinese Core, CSCD Expanded, EI indexed)

J. Wang, X. Zhang, X. Ma, and **Y. Gao**. “An Improved Model-Free Adaptive Control Algorithm with Differential Element.” *Proceedings of the 2022 China Automation Congress (CAC)*, pp. 684–689, 2022.

X. Guo, M. Xiao, **Y. Gao**, Q. Wang, and Y. Wan. “Fix-Frequency Robust Power Model Predictive Control Method for Three-Phase PWM Rectifiers under Unbalanced Grid Conditions.” *Journal of Power Electronics*, vol. 20, no. 5, pp. 1283–1294, 2020. (SCI/EI indexed)

ADDITIONAL LEADERSHIP EXPERIENCE

Guangzhou Qiwu Intelligent Technology Co., Ltd. 2023 – 2024 (**Co-Founder**)

Led cross-functional collaboration on AI-related product development and deployment, gaining experience in technical leadership, team coordination, and prototype-to-delivery execution.

SELECTED TECHNICAL KEYWORDS

Marine robotics; underwater autonomy; disturbance-aware planning; motion planning; trajectory optimization; robust perception; visual localization; state estimation; over-actuated ROVs; ROS/ROS2; real-robot validation; adaptive marine sensing.

REFERENCES

Dr. Yi Cai

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Dr. Jun Ma

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